

Thesis-Level Derivation of the Inner Current Loop and Outer Voltage Loop for Average-Current-Mode CCM PFC

FAN9672 Two-Channel Interleaved Boost PFC Controller

Laplace-Domain Derivation, Physical Interpretation, and Design Mapping

Prepared for the 2-Channel Interleaved CCM PFC Design Study

Output Bus: 393.7 V, Low-Line Power: 1700 W, High-Line Power: 3600 W

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Executive Summary

This document develops the control-loop theory for a two-channel interleaved continuous-conduction-mode (CCM) boost power-factor-correction (PFC) converter using the FAN9672 average-current-mode controller. The goal is to derive the inner current-loop and outer voltage-loop equations from the averaged boost model, carry the equations into the Laplace domain, identify every physical term, and show how the final open-loop gains are assembled.

The derivation follows the structure of the user-provided current-loop and voltage-loop derivation notes and the Chapter 6 control-scheme summary. The source notes define the cascaded current and voltage loop architecture, the full duty-to-current plant, and the voltage-loop energy-balance model. The FAN9672 data sheet identifies the device as a two-channel interleaved CCM PFC controller with average-current-mode control, differential current sensing, brown-in/out protection, and the relevant pins for IAC, RLPK, GC, LS, VEA, IEA, FBPF, and current sense. AN4165D and AND9925D provide the FAN967x application context and gain-modulator/support-pin design equations, while AN5257 provides an independent average-current-mode interleaved-PFC control-theory reference.

Important. A transfer function cannot be obtained without a modelling basis. Therefore the phrase “no assumption and no approximation” is interpreted here as follows: after selecting the CCM averaged boost model and applying first-order small-signal linearization, no model terms are dropped. The derivations retain inductor DCR r_L , capacitor ESR r_C , finite load resistance R_{LOAD} , duty-dependent terms, current-sense filtering, closed-current-loop tracking, RHP zero, and ESR zero. Numerical results are rounded only for presentation.

The final inner current-loop gain is

$$T_i(s) = G_{id}(s) \frac{R_{CS}}{V_{RAMP}} H_{CS}(s) G_{mi}(s). \quad (1)$$

The final outer voltage-loop gain is

$$T_v(s) = \frac{K_{MAX} I_{OUT}}{5} \frac{V_{FBPF}}{V_{OUT}} \frac{T_i(s)/N_{CH}}{1 + T_i(s)/N_{CH}} \frac{(1 + sC_{OrC})(1 - s/\omega_{RHP})}{C_{Os} + 2/R_{LOAD}} G_{MV} Z_{comp}(s). \quad (2)$$

1 Reference Basis and Scope

1.1 Controller and control method

The FAN9672 is a two-channel interleaved CCM PFC controller intended for PFC pre-regulators. The data sheet describes average-current-mode operation, two-channel interleaving, differential current sensing, brown-in/out protection, programmable frequency, output-voltage feedback through FBPF, current error-amplifier outputs IEA1/IEA2, and gate-drive outputs OPFC1/OPFC2. Those functions correspond directly to the two-loop architecture used in this derivation.

The supporting reference documents establish the following design context:

- The Chapter 6 control-scheme summary states that ACM PFC uses a fast inner current loop and a slow outer voltage loop, then lists the open-loop forms $T_i(s)$ and $T_v(s)$.
- The inner-loop theory note derives the full $G_{id}(s)$ from the averaged CCM boost model while retaining r_L , r_C , R_{LOAD} , L , C , and duty-dependent terms.
- The outer-loop theory note derives the voltage-loop plant from energy balance and explains why the denominator is $C_O s + 2/R_{LOAD}$ rather than $C_O s + 1/R_{LOAD}$.
- AN5257 provides an independent interleaved-PFC control theory reference using average-current-mode control, current loops, voltage loop, and feedforward concepts.
- FAN9672-D, AN4165D, and AND9925D connect the theory to FAN967x pins and internal functions such as IAC, RLPK, VIR, GC, LS, GMI, GMV, IEA, VEA, BIBO, and current-limit blocks.

1.2 Design constants used for numerical checks

The symbolic derivation is general. The numerical examples use the current design values shown in Table 1.

Table 1: Design constants used for numerical evaluation

Parameter	Symbol	Value
Output voltage	V_{OUT}	393.7 V
Low-line output power	$P_{OUT,LL}$	1700 W
High-line output power	$P_{OUT,HL}$	3600 W
Number of interleaved channels	N_{CH}	2
PFC inductor per phase	L_ϕ	235 μ H
Output capacitance	C_O	2200 μ F
Inductor DCR	r_L	10 m Ω
Output capacitor ESR	r_C	10 m Ω
Current-sense resistor	R_{CS}	15 m Ω
PWM ramp voltage	V_{RAMP}	5 V
Current OTA transconductance	G_{MI}	88 μ S
Voltage OTA transconductance	G_{MV}	100 μ S
Multiplier power factor	K_{MAX}	1.4
Current-loop crossover target	f_{ci}	8 kHz
Voltage-loop crossover target	f_{cv}	17 Hz
Sense filter	R_f, C_f	2 k Ω , 470 pF

2 Average-Current-Mode PFC Control Architecture

2.1 Nested-loop structure

Average-current-mode PFC is a nested two-loop system. The inner current loop is fast and forces the boost inductor current to track a current command proportional to the rectified input voltage. The outer voltage loop is slow and adjusts only the current-command amplitude to regulate the DC bus.

For each phase, the current path is

$$\hat{d}(s) \longrightarrow \hat{i}_L(s) \longrightarrow \hat{v}_{CS}(s) \longrightarrow \hat{v}_{IEA}(s) \longrightarrow \hat{d}(s). \quad (3)$$

For the voltage loop, the path is

$$\hat{v}_o(s) \longrightarrow \hat{v}_{FB}(s) \longrightarrow \hat{v}_{VEA}(s) \longrightarrow \hat{i}_{cmd}(s) \longrightarrow \hat{p}_{in}(s) \longrightarrow \hat{v}_o(s). \quad (4)$$

The loop separation is intentional:

$$f_{cv} \ll 2f_{line} \ll f_{ci} \ll f_{sw}. \quad (5)$$

For this design,

$$17 \text{ Hz} \ll 120 \text{ Hz} \ll 8 \text{ kHz} \ll 70 \text{ kHz}. \quad (6)$$

2.2 Why the current loop is closed first

The current loop must be designed first because the voltage loop commands current. In the voltage-loop equation, the inner loop appears as the tracking term

$$G_{i,cl}(s) = \frac{T_i(s)/N_{CH}}{1 + T_i(s)/N_{CH}}. \quad (7)$$

At frequencies much lower than the current-loop crossover, this term is close to unity. It is nevertheless retained in the final equation because the request is to keep the full loop structure and avoid replacing a dynamic block by a constant.

3 Mathematical Basis: Averaging, Perturbation, and Laplace Transform

3.1 Averaged large-signal modelling

For a CCM boost phase, the switch ON and OFF intervals can be duty-cycle averaged over one switching period. The averaged equations are continuous-time equations in the slow variables i_L , v_C , and v_o .

The selected model retains:

- inductor DCR r_L ,
- output capacitor ESR r_C ,
- finite load resistance $R = R_{LOAD}$,
- duty D and duty complement $D' = 1 - D$,
- output-capacitor dynamics,
- boost output-current coupling through $D'i_L$.

3.2 Small-signal perturbation

Each large-signal variable is written as a steady-state value plus a small perturbation:

$$\begin{aligned} i_L &= I_L + \hat{i}_L, \\ v_C &= V_C + \hat{v}_C, \\ v_o &= V_O + \hat{v}_o, \\ d &= D + \hat{d}, \\ D' &= 1 - D - \hat{d}. \end{aligned} \tag{8}$$

The product $D'i_L$ becomes

$$\begin{aligned} D'i_L &= (1 - D - \hat{d})(I_L + \hat{i}_L) \\ &= (1 - D)I_L + (1 - D)\hat{i}_L - I_L\hat{d} - \hat{d}\hat{i}_L. \end{aligned} \tag{9}$$

The first-order perturbation is therefore

$$\widehat{D'i_L} = D'\hat{i}_L - I_L\hat{d}. \tag{10}$$

The term $\hat{d}\hat{i}_L$ is a second-order perturbation. Removing second-order perturbation products is not an optional engineering approximation; it is the definition of a first-order small-signal transfer function.

3.3 Laplace transform convention

All perturbation variables are transformed using zero initial conditions. Therefore

$$\mathcal{L}\left\{\frac{d\hat{x}}{dt}\right\} = s\hat{x}(s). \tag{11}$$

For readability, hats are kept on small-signal variables but the argument (s) is sometimes omitted inside longer derivations.

4 Inner Current-Loop Theory Derivation

4.1 Definition of the current-loop plant

The per-phase duty-to-inductor-current plant is

$$G_{id}(s) = \frac{\hat{i}_L(s)}{\hat{d}(s)}. \quad (12)$$

The current loop also contains current sensing, a current-sense filter, current OTA compensation, and PWM ramp gain. The final current-loop gain is

$$T_i(s) = G_{id}(s) R_{CS} H_{CS}(s) G_{mi}(s) F_m, \quad F_m = \frac{1}{V_{RAMP}}. \quad (13)$$

Rearranging,

$$T_i(s) = G_{id}(s) \frac{R_{CS}}{V_{RAMP}} H_{CS}(s) G_{mi}(s). \quad (14)$$

4.2 Large-signal averaged boost equations

For one boost phase,

$$L \frac{di_L}{dt} = v_{in} - r_L i_L - D' v_o. \quad (15)$$

The output current delivered by the boost diode path is

$$i_d = D' i_L. \quad (16)$$

Including capacitor ESR, the capacitor state equation can be written as

$$C(R + r_C) \frac{dv_C}{dt} = R D' i_L - v_C. \quad (17)$$

The terminal output voltage is

$$v_o = \frac{R}{R + r_C} v_C + \frac{R r_C}{R + r_C} D' i_L. \quad (18)$$

Equations (15)–(18) define the full averaged model used for the inner-loop plant derivation.

4.3 Small-signal capacitor equation

Apply Eq. (10) to Eq. (17):

$$C(R + r_C) \frac{d\hat{v}_C}{dt} = R \left(D' \hat{i}_L - I_L \hat{d} \right) - \hat{v}_C. \quad (19)$$

Taking the Laplace transform,

$$C(R + r_C) s \hat{v}_C = R D' \hat{i}_L - R I_L \hat{d} - \hat{v}_C. \quad (20)$$

Collecting $\hat{v}_C$ terms,

$$[1 + sC(R + r_C)] \hat{v}_C = R \left(D' \hat{i}_L - I_L \hat{d} \right). \quad (21)$$

Therefore

$$\hat{v}_C(s) = \frac{R \left(D' \hat{i}_L(s) - I_L \hat{d}(s) \right)}{1 + sC(R + r_C)}. \quad (22)$$

4.4 Small-signal output-voltage equation

From Eq. (18),

$$\hat{v}_o = \frac{R}{R+r_C} \hat{v}_C + \frac{Rr_C}{R+r_C} (D'\hat{i}_L - I_L\hat{d}). \quad (23)$$

Substitute Eq. (22):

$$\begin{aligned} \hat{v}_o &= \frac{R}{R+r_C} \frac{R(D'\hat{i}_L - I_L\hat{d})}{1+sC(R+r_C)} + \frac{Rr_C}{R+r_C} (D'\hat{i}_L - I_L\hat{d}) \\ &= \frac{R}{R+r_C} \left[\frac{R}{1+sC(R+r_C)} + r_C \right] (D'\hat{i}_L - I_L\hat{d}). \end{aligned} \quad (24)$$

Put the bracket over a common denominator:

$$\frac{R}{1+sC(R+r_C)} + r_C = \frac{R+r_C+sCr_C(R+r_C)}{1+sC(R+r_C)}. \quad (25)$$

Since

$$R+r_C+sCr_C(R+r_C) = (R+r_C)(1+sCr_C), \quad (26)$$

then

$$\hat{v}_o = R \frac{1+sCr_C}{1+sC(R+r_C)} (D'\hat{i}_L - I_L\hat{d}). \quad (27)$$

4.5 Small-signal inductor equation and control-to-current plant

For the control-to-current transfer function, set the input-voltage perturbation to zero, $\hat{v}_{in} = 0$. From Eq. (15),

$$L \frac{d\hat{i}_L}{dt} = -r_L \hat{i}_L - D' \hat{v}_o + V_O \hat{d}. \quad (28)$$

Laplace transform gives

$$(Ls + r_L) \hat{i}_L + D' \hat{v}_o = V_O \hat{d}. \quad (29)$$

Substitute Eq. (27):

$$(Ls + r_L) \hat{i}_L + D' R \frac{1+sCr_C}{1+sC(R+r_C)} (D'\hat{i}_L - I_L\hat{d}) = V_O \hat{d}. \quad (30)$$

Multiply by $1 + sC(R+r_C)$:

$$\begin{aligned} &[(Ls + r_L)(1 + sC(R+r_C)) + D'^2 R(1 + sCr_C)] \hat{i}_L \\ &= [V_O(1 + sC(R+r_C)) + D' R I_L(1 + sCr_C)] \hat{d}. \end{aligned} \quad (31)$$

Thus the exact small-signal duty-to-current plant before steady-state substitution is

$$G_{id}(s) = \frac{V_O(1 + sC(R+r_C)) + D' R I_L(1 + sCr_C)}{(Ls + r_L)(1 + sC(R+r_C)) + D'^2 R(1 + sCr_C)}. \quad (32)$$

4.6 Steady-state substitutions and numerator zero

At the operating point,

$$I_O = \frac{V_O}{R}, \quad D' I_L = I_O = \frac{V_O}{R}. \quad (33)$$

Therefore,

$$D'RI_L = V_O. \quad (34)$$

Substitute Eq. (34) into the numerator of Eq. (32):

$$\begin{aligned} N(s) &= V_O(1 + sC(R + r_C)) + V_O(1 + sCr_C) \\ &= V_O [2 + sC(R + r_C) + sCr_C] \\ &= V_O [2 + sC(R + 2r_C)]. \end{aligned} \quad (35)$$

Now write

$$2 + sC(R + 2r_C) = C(R + 2r_C) \left(s + \frac{2}{C(R + 2r_C)} \right). \quad (36)$$

Since

$$\frac{2}{C(R + 2r_C)} = \frac{1}{C(R/2 + r_C)}, \quad (37)$$

the numerator zero is

$$\omega_z = \frac{1}{C(R/2 + r_C)}. \quad (38)$$

This zero is not the capacitor ESR zero $1/(Cr_C)$, because it contains $R/2 + r_C$.

4.7 Denominator expansion

The denominator of Eq. (32) is

$$D_{den}(s) = (Ls + r_L)(1 + sC(R + r_C)) + D'^2R(1 + sCr_C). \quad (39)$$

Expand:

$$\begin{aligned} D_{den}(s) &= Ls + r_L + s^2LC(R + r_C) + sr_LC(R + r_C) \\ &\quad + D'^2R + sD'^2RCr_C. \end{aligned} \quad (40)$$

Collect powers of s :

$$D_{den}(s) = s^2LC(R + r_C) + s [L + r_LC(R + r_C) + D'^2RCr_C] + [r_L + D'^2R]. \quad (41)$$

Divide by $LC(R + r_C)$ to obtain

$$D_{norm}(s) = s^2 + a_1s + a_0, \quad (42)$$

where

$$a_1 = \frac{C [r_L(R + r_C) + Rr_C(1 - D)^2] + L}{LC(R + r_C)} \quad (43)$$

and

$$a_0 = \frac{(1 - D)^2R + r_L}{LC(R + r_C)}. \quad (44)$$

4.8 Final full duty-to-current plant

Combining the normalized numerator and denominator gives

$$G_{id}(s) = \frac{V_{OUT}}{L} \frac{R_{LOAD} + 2r_C}{R_{LOAD} + r_C} \frac{s + \omega_z}{s^2 + a_1s + a_0} \quad (45)$$

with

$$\omega_z = \frac{1}{C_O(R_{LOAD}/2 + r_C)}. \quad (46)$$

The natural frequency and quality factor of the second-order denominator are

$$\omega_0 = \sqrt{a_0}, \quad f_0 = \frac{\sqrt{a_0}}{2\pi}, \quad Q = \frac{\sqrt{a_0}}{a_1}. \quad (47)$$

4.9 Current-sense filter and PWM ramp gain

The current-sense filter is

$$H_{CS}(s) = \frac{1}{1 + s/\omega_{RC}}, \quad \omega_{RC} = \frac{1}{R_f C_f}. \quad (48)$$

For $R_f = 2 \text{ k}\Omega$ and $C_f = 470 \text{ pF}$,

$$f_{RC} = \frac{1}{2\pi R_f C_f} = 169.3 \text{ kHz}. \quad (49)$$

The PWM ramp gain is

$$F_m = \frac{1}{V_{RAMP}}, \quad (50)$$

and the current-sense/ramp normalization is

$$\frac{R_{CS}}{V_{RAMP}} = \frac{0.015}{5} = 0.003. \quad (51)$$

4.10 Type-II current OTA compensator

The FAN9672 current amplifier is an OTA. The transconductance amplifier converts error voltage into output current, and the external impedance $Z_{IEA}(s)$ converts that current into the compensation voltage:

$$G_{mi}(s) = G_{MI} Z_{IEA}(s). \quad (52)$$

For a Type-II compensation shape,

$$Z_{IEA}(s) = R_{IC} \frac{1 + \omega_{zi}/s}{1 + s/\omega_{pi}}. \quad (53)$$

This form provides high low-frequency gain, a zero for phase boost near crossover, and a high-frequency pole for noise-gain control.

4.11 Final current-loop gain

Substitute Eq. (45), Eq. (48), and Eq. (53) into Eq. (14):

$$T_i(s) = \frac{V_{OUT}}{L} \frac{R_{LOAD} + 2r_C}{R_{LOAD} + r_C} \frac{s + \omega_z}{s^2 + a_1 s + a_0} \frac{R_{CS}}{V_{RAMP}} G_{MI} R_{IC} \frac{1 + \omega_{zi}/s}{1 + s/\omega_{pi}} \frac{1}{1 + s/\omega_{RC}} \quad (54)$$

5 Inner Current-Loop Numerical Check

5.1 Operating-point table

Table 2 evaluates the current-loop plant parameters using the full expressions above. The RHP-zero value shown is listed for boost-voltage-loop awareness; the RHP zero belongs to the duty-to-output-voltage path, not to the duty-to-inductor-current plant used for the current loop.

Table 2: Operating-point and uncompensated current-loop quantities at 8 kHz

V_{AC} (V)	P_{OUT} (W)	$V_{in,pk}$ (V)	D –	R_{LOAD} (Ω)	F_0 (Hz)	Q –	f_{RHP} (kHz)	$T_{i,unc}$ (dB)	$\angle T_{i,unc}$ (deg)
90	1700	127.28	0.67671	91.176	71.59	8.653	12.908	-20.009	-92.66
110	1700	155.56	0.60487	91.176	87.49	10.146	19.282	-20.009	-92.65
120	1700	169.71	0.56895	91.176	95.44	10.815	22.947	-20.009	-92.65
132	1700	186.68	0.52584	91.176	104.97	11.550	27.766	-20.008	-92.65
180	3600	254.56	0.35342	43.055	143.14	12.686	24.381	-20.006	-92.65
210	3600	296.98	0.24566	43.055	166.99	13.570	33.185	-20.005	-92.64
220	3600	311.13	0.20974	43.055	174.94	13.795	36.421	-20.005	-92.64
230	3600	325.27	0.17381	43.055	182.88	13.988	39.808	-20.004	-92.64
264	3600	373.35	0.05168	43.055	209.91	14.435	52.447	-20.003	-92.63

5.2 Current-loop Bode plot

Figure 1 shows the compensated open current-loop gain using the derived $G_{id}(s)$ and the Type-II compensator.

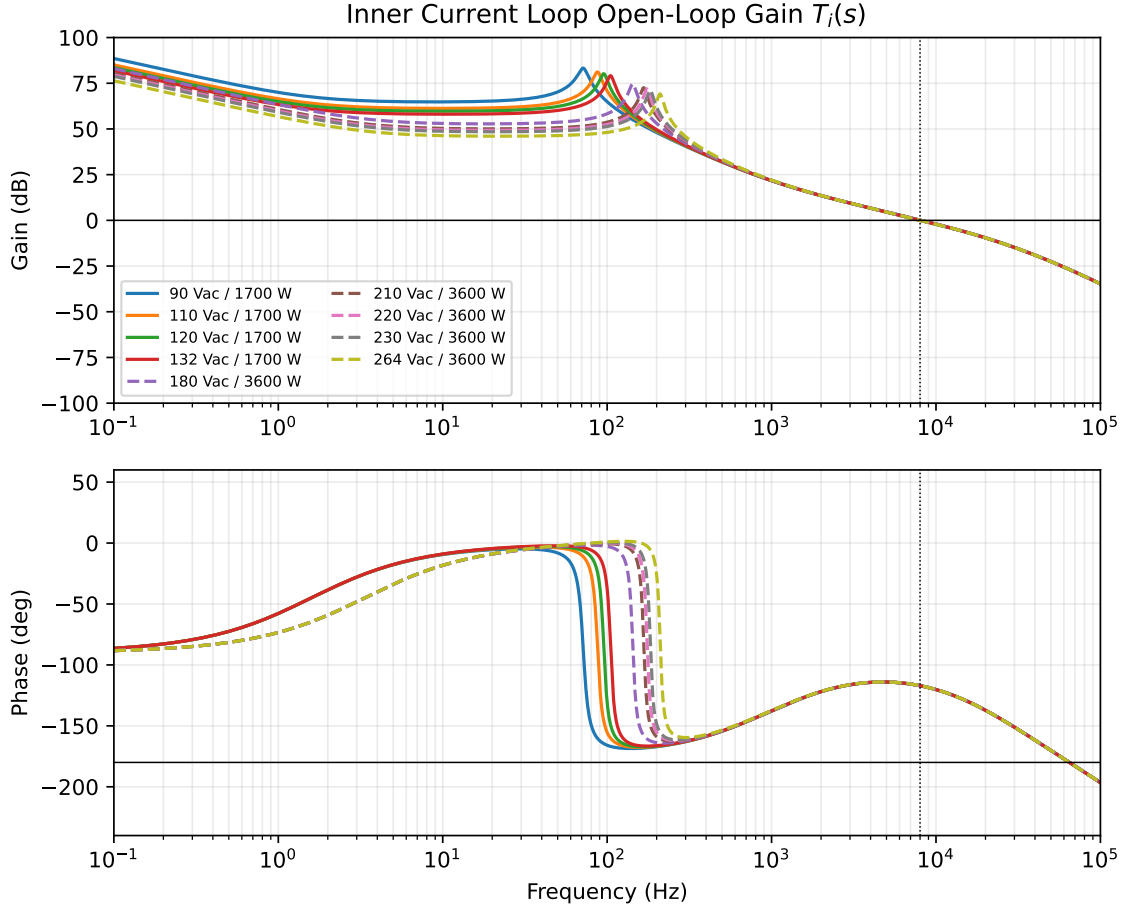


Figure 1: Inner current-loop open-loop Bode plot using the full duty-to-current plant, current-sense filter, ramp gain, and Type-II OTA compensation.

6 Outer Voltage-Loop Theory Derivation

6.1 What the outer loop controls

The voltage loop regulates the DC bus. In a PFC converter it does not directly command duty. Instead, it commands the amplitude of the sinusoidal current reference. The inner current loop then makes the inductor current track that reference.

The final voltage-loop gain is a product of five blocks:

$$T_v(s) = G_{MOD} G_{i,cl}(s) G_{vp}(s) H_v G_{mv}(s). \quad (55)$$

Each block has a separate physical origin:

Table 3: Outer voltage-loop blocks

Block	Expression	Physical meaning
Multiplier / gain modulator	G_{MOD}	VEA command to current-command amplitude
Closed current-loop tracking	$G_{i,cl}(s)$	Accuracy of current-reference tracking
Output plant	$G_{vp}(s)$	Output power/current to bus voltage
Feedback divider	H_v	Output voltage to FBPFC voltage
Voltage compensator	$G_{mv}(s)$	Voltage OTA and compensation network

6.2 Energy-balance output plant

The output capacitor stores energy

$$E_C = \frac{1}{2} C_O V_o^2. \quad (56)$$

The rate of change of stored energy equals input power minus load power:

$$C_O V_o \frac{dV_o}{dt} = p_{in}(t) - \frac{V_o^2}{R_{LOAD}}. \quad (57)$$

Let

$$V_o = V_{OUT} + \hat{v}_o, \quad p_{in} = P_{IN} + \hat{p}_{in}. \quad (58)$$

At steady state,

$$P_{IN} = \frac{V_{OUT}^2}{R_{LOAD}}. \quad (59)$$

The load power perturbation is

$$\begin{aligned} \frac{V_o^2}{R_{LOAD}} &= \frac{(V_{OUT} + \hat{v}_o)^2}{R_{LOAD}} \\ &= \frac{V_{OUT}^2}{R_{LOAD}} + \frac{2V_{OUT}}{R_{LOAD}} \hat{v}_o + \frac{\hat{v}_o^2}{R_{LOAD}}. \end{aligned} \quad (60)$$

The first-order perturbation is

$$\widehat{p_{load}} = \frac{2V_{OUT}}{R_{LOAD}} \hat{v}_o. \quad (61)$$

The left-hand side of Eq. (57) becomes

$$C_O (V_{OUT} + \hat{v}_o) \frac{d\hat{v}_o}{dt}. \quad (62)$$

The first-order term is

$$C_O V_{OUT} \frac{d\hat{v}_o}{dt}. \quad (63)$$

Therefore,

$$C_O V_{OUT} \frac{d\hat{v}_o}{dt} = \hat{p}_{in} - \frac{2V_{OUT}}{R_{LOAD}} \hat{v}_o. \quad (64)$$

Divide by V_{OUT} :

$$C_O \frac{d\hat{v}_o}{dt} + \frac{2}{R_{LOAD}} \hat{v}_o = \frac{\hat{p}_{in}}{V_{OUT}}. \quad (65)$$

Laplace transform gives

$$\left(C_O s + \frac{2}{R_{LOAD}} \right) \hat{v}_o(s) = \frac{\hat{p}_{in}(s)}{V_{OUT}}. \quad (66)$$

Thus

$$\frac{\hat{v}_o(s)}{\hat{p}_{in}(s)} = \frac{1}{V_{OUT}} \frac{1}{C_O s + 2/R_{LOAD}}. \quad (67)$$

If the command variable is an equivalent output current perturbation $\hat{i}_{out,eq} = \hat{p}_{in}/V_{OUT}$, then

$$\frac{\hat{v}_o(s)}{\hat{i}_{out,eq}(s)} = \frac{1}{C_O s + 2/R_{LOAD}}. \quad (68)$$

This is why the voltage-loop denominator is $C_O s + 2/R_{LOAD}$.

6.3 Output capacitor ESR zero

The terminal output voltage is

$$v_o = v_C + r_C i_C. \quad (69)$$

With

$$i_C = C_O \frac{dv_C}{dt}, \quad (70)$$

Laplace transformation gives

$$v_o(s) = v_C(s) + r_C C_O s v_C(s) = v_C(s)(1 + s C_O r_C). \quad (71)$$

Therefore the output plant contains the ESR-zero factor

$$G_{ESR}(s) = 1 + s C_O r_C, \quad \omega_{ESR} = \frac{1}{C_O r_C}. \quad (72)$$

6.4 Boost right-half-plane zero

A CCM boost converter has non-minimum-phase output dynamics. Increasing duty reduces the instantaneous off-time $D' = 1 - D$ and initially reduces energy delivery to the output before the inductor current rises. This creates a right-half-plane zero in the output-voltage path.

For the interleaved output plant written with the effective inductance appropriate to total output energy transfer,

$$\omega_{RHP} = \frac{R_{LOAD}(1 - D)^2}{L_{eq}}, \quad L_{eq} = \frac{L_\phi}{N_{CH}}. \quad (73)$$

The RHP zero appears as

$$1 - \frac{s}{\omega_{RHP}}. \quad (74)$$

The minus sign is essential: it contributes phase lag rather than phase lead.

6.5 Full voltage power-stage plant

Combining the energy-balance denominator, ESR zero, and RHP zero gives

$$G_{vp}(s) = \frac{(1 + sC_{OC}r_C)(1 - s/\omega_{RHP})}{C_O s + 2/R_{LOAD}} \quad (75)$$

6.6 Closed inner current-loop tracking

The voltage loop commands current, but the current does not track perfectly. The closed current-loop tracking term is retained as

$$G_{i,cl}(s) = \frac{T_i(s)/N_{CH}}{1 + T_i(s)/N_{CH}}. \quad (76)$$

If $|T_i(s)/N_{CH}| \gg 1$, this term approaches one. The full expression is kept because this document does not replace dynamic blocks by constants in the final loop equation.

6.7 Feedback divider gain

The feedback divider gives

$$H_v = \frac{R_{BOT}}{R_{TOP} + R_{BOT}}. \quad (77)$$

At regulation,

$$H_v = \frac{V_{FBPFC}}{V_{OUT}}. \quad (78)$$

Using $V_{FBPFC} = 2.5 \text{ V}$ and $V_{OUT} = 393.7 \text{ V}$,

$$H_v = \frac{2.5}{393.7} = 0.006350. \quad (79)$$

6.8 Gain modulator block

For voltage-loop stability analysis, the system-level averaged command gain is

$$G_{MOD} = \frac{K_{MAX} I_{OUT}}{5} = \frac{K_{MAX} P_{OUT}}{5 V_{OUT}}. \quad (80)$$

This is the line-cycle averaged loop-analysis form. FAN9672/AND9925 hardware equations for IAC, RLPK, LPK, VIR, and gain-modulator internals are used to design support-pin scaling and power-command limits; they should not be directly inserted into Eq. (55) as though they were the averaged closed outer-loop gain at every line angle.

For the two power levels:

$$\begin{aligned} G_{MOD,1700} &= \frac{1.4(1700/393.7)}{5} = 1.209 \text{ A/V}, \\ G_{MOD,3600} &= \frac{1.4(3600/393.7)}{5} = 2.561 \text{ A/V}. \end{aligned} \quad (81)$$

6.9 Voltage OTA compensator

The voltage OTA produces output current

$$i_{OTA} = G_{MV}v_{err}. \quad (82)$$

The compensation impedance converts this current to a voltage-loop command:

$$v_c = i_{OTA}Z_{comp}(s). \quad (83)$$

Therefore,

$$G_{mv}(s) = G_{MV}Z_{comp}(s). \quad (84)$$

For a Type-III target shape,

$$Z_{comp}(s) = K_v \frac{(1 + s/\omega_{z1})(1 + s/\omega_{z2})}{s(1 + s/\omega_{p1})(1 + s/\omega_{p2})}. \quad (85)$$

The selected target locations are $f_{z1} = 3$ Hz, $f_{z2} = 12$ Hz, $f_{p1} = 17$ Hz, and $f_{p2} = 50$ Hz.

6.10 Final voltage-loop gain

Substitute Eq. (80), Eq. (76), Eq. (75), Eq. (78), and Eq. (84) into Eq. (55):

$$T_v(s) = \frac{K_{MAX}I_{OUT}}{5} \frac{V_{FBPFC}}{V_{OUT}} \frac{T_i(s)/N_{CH}}{1 + T_i(s)/N_{CH}} \frac{(1 + sC_{OrC})(1 - s/\omega_{RHP})}{C_{Os} + 2/R_{LOAD}} G_{MV}Z_{comp}(s) \quad (86)$$

7 Outer Voltage-Loop Numerical Check

7.1 Voltage-loop Bode plot

Figure 2 shows representative voltage-loop open-loop gain using the full voltage-loop expression and the final Type-III target network.

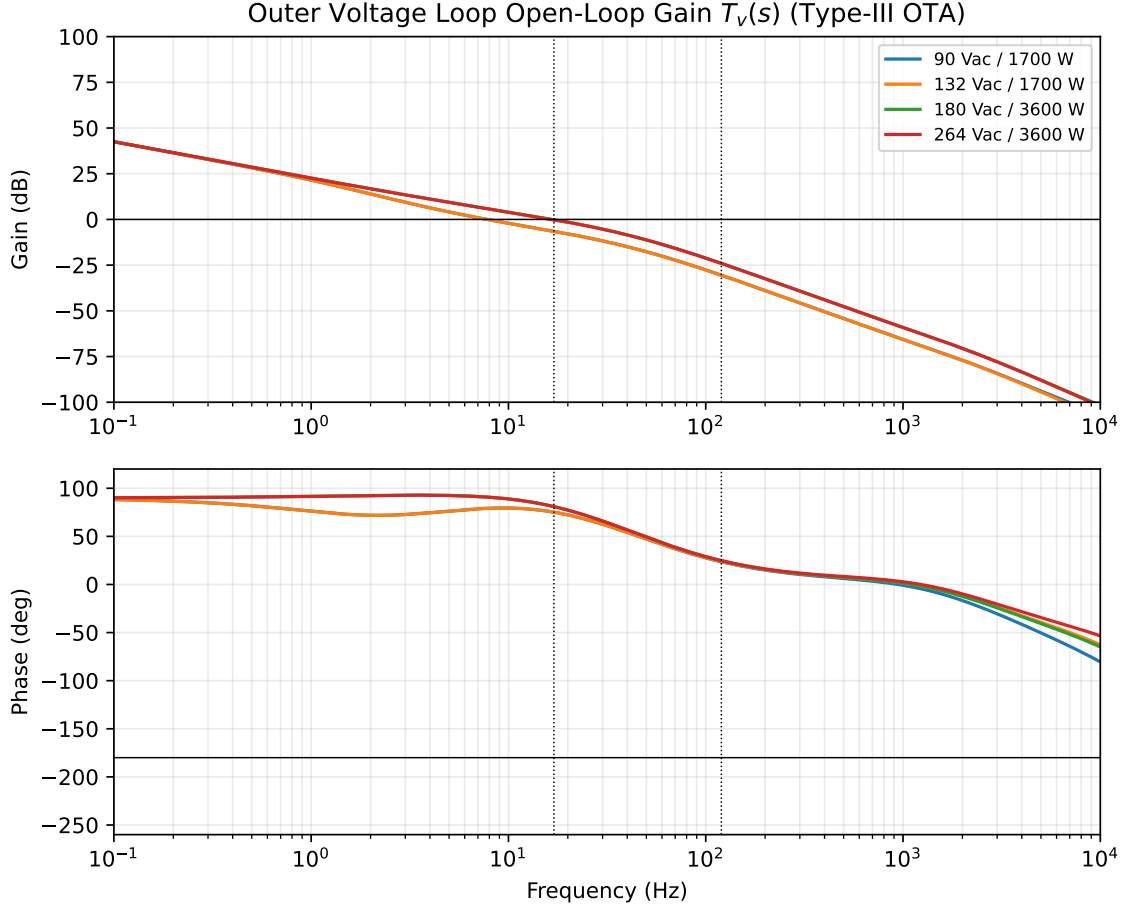


Figure 2: Outer voltage-loop open-loop Bode plot using the energy-balance plant, RHP zero, ESR zero, closed current-loop tracking term, multiplier gain, feedback divider, and Type-III voltage OTA compensation.

7.2 Voltage-loop gain at 17 Hz

Table 4: Representative voltage-loop gain at 17 Hz with the selected Type-III network

V_{AC}	P_{OUT}	$ T_v(j2\pi 17) $ (dB)	$\angle T_v$ (deg)
90	1700	-6.46	75.14
110	1700	-6.46	75.17
120	1700	-6.47	75.17
132	1700	-6.47	75.18
180	3600	-0.10	80.98
210	3600	-0.11	80.98
220	3600	-0.12	80.97
230	3600	-0.13	80.97
264	3600	-0.15	80.95

8 Relationship Between Inner and Outer Loop

The inner loop shapes the current-control dynamics and determines how accurately the inductor current follows the command. The outer loop uses the closed inner loop as an actuator. This relationship is exactly captured by

$$G_{i,cl}(s) = \frac{T_i(s)/N_{CH}}{1 + T_i(s)/N_{CH}}. \quad (87)$$

At the voltage-loop crossover of 17 Hz, the current-loop bandwidth is much higher, so $G_{i,cl}(s)$ is close to one. Nevertheless, it remains in the complete symbolic equation. If the current loop were poorly tuned, too slow, saturated, or phase-shifted, the voltage loop would see that nonideal behavior through this term.

9 120 Hz Rejection and THD Tradeoff

For a 60 Hz line, the rectified input power contains a twice-line component at 120 Hz. The output capacitor must absorb this energy pulsation. If the voltage loop has excessive gain at 120 Hz, it will modulate the current-command amplitude at 120 Hz. That modulation distorts the input-current waveform and degrades THD.

The outer-loop design therefore trades off:

- low crossover frequency for better 120 Hz rejection and lower THD,
- higher crossover frequency for faster load-transient response,
- Type-III phase boost for phase margin near 17 Hz,
- high-frequency pole placement for reduced ripple/noise injection into the current command.

A 17 Hz crossover is below 120 Hz by approximately a factor of seven. This is a deliberate ACM PFC design choice.

10 FAN9672 Controller Mapping

Table 5: Mapping from derived control blocks to FAN9672 functions

Derived block	FAN9672 implementation	Comments
$G_{id}(s)$	External boost power stage	Set by L , C_O , r_L , r_C , R_{LOAD} , D
R_{CS}	CS1+/CS1-, CS2+/CS2-	Differential current sensing
$H_{CS}(s)$	External sense filter	$R_f = 2\text{ k}\Omega$, $C_f = 470\text{ pF}$
$G_{mi}(s)$	GMI + IEA1/IEA2 network	Current Type-II compensation
F_m	Internal PWM ramp	$1/V_{RAMP}$
G_{MOD}	IAC/RLPK/VIR/LPK/gain modulator	Use averaged form for loop analysis
H_v	FBPFC divider	V_{FBPFC}/V_{OUT}
$G_{mv}(s)$	GMV + VEA network	Voltage Type-II/Type-III compensation
BIBO	Brown-in/out pin network	Input UVP/brownout
GC, LS	LPT/gain-control support pins	Hardware support functions

11 Thesis-Level Conclusions

The inner current-loop equation is derived from the full averaged CCM boost model by retaining the inductor DCR, capacitor ESR, finite load, duty complement, and dynamic output coupling. The final plant is not the simple V/Ls approximation; it is a second-order plant with a numerator zero at $1/[C_O(R_{LOAD}/2 + r_C)]$.

The outer voltage-loop equation is derived from energy balance rather than simple capacitor current balance. This is why its denominator is $C_Os + 2/R_{LOAD}$. The factor of two is physically caused by the derivative of load power with respect to voltage. The outer-loop plant also includes the capacitor ESR zero and the boost RHP zero.

For FAN9672 average-current-mode PFC, the loop equations must keep the controller-specific blocks: R_{CS}/V_{RAMP} , $G_{MI}Z_{IEA}(s)$, $G_{MV}Z_{comp}(s)$, gain modulator $K_{MAX}I_{OUT}/5$, divider gain V_{FBPFC}/V_{OUT} , and closed current-loop tracking $[T_i/N_{CH}]/[1 + T_i/N_{CH}]$.

The final equations are:

$$T_i(s) = \frac{V_{OUT}}{L} \frac{R_{LOAD} + 2r_C}{R_{LOAD} + r_C} \frac{s + \omega_z}{s^2 + a_1s + a_0} \frac{R_{CS}}{V_{RAMP}} G_{MI}Z_{IEA}(s) \frac{1}{1 + s/\omega_{RC}}. \quad (88)$$

$$T_v(s) = \frac{K_{MAX}I_{OUT}}{5} \frac{V_{FBPFC}}{V_{OUT}} \frac{T_i(s)/N_{CH}}{1 + T_i(s)/N_{CH}} \frac{(1 + sC_{OrC})(1 - s/\omega_{RHP})}{C_Os + 2/R_{LOAD}} G_{MV}Z_{comp}(s). \quad (89)$$

A Appendix A: Symbol Glossary

Symbol	Meaning
D	Steady-state duty ratio
$D' = 1 - D$	Duty complement / boost diode conduction fraction
L or L_ϕ	Per-phase boost inductance
C_O	Total output capacitance
r_L	Inductor DCR
r_C	Output capacitor ESR
R_{LOAD}	Equivalent output load resistance, V_{OUT}^2/P_{OUT}
$G_{id}(s)$	Duty-to-inductor-current plant
$H_{CS}(s)$	Current-sense filter
$G_{mi}(s)$	Current OTA compensation block
$G_{mv}(s)$	Voltage OTA compensation block
G_{MOD}	Voltage-loop command to current-command gain
$G_{vp}(s)$	Output voltage/power plant
$G_{i,cl}(s)$	Closed current-loop tracking term
$T_i(s)$	Inner current-loop open-loop gain
$T_v(s)$	Outer voltage-loop open-loop gain

B Appendix B: Source Notes and References

1. onsemi, *FAN9672/D Two-Channel Interleaved CCM PFC Controller*, data sheet, Rev. 6, June 2023.
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3. onsemi, *AND9925/D FAN9672/FAN9673 Tips and Tricks*, application note, Rev. 3, May 2024.
4. NXP/Freescale, *AN5257 Average Current Mode Interleaved PFC Control*, Rev. 0, February 2016.
5. User-provided document, *Inner Current Loop Theory Derivation*, attached Word document.
6. User-provided document, *Outer Voltage Loop Theory Derivation*, attached Word document.
7. User-provided document, *Chapter 6 Control Scheme*, attached PDF.